

Probability

GCSE Maths · Worked Solutions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

A fair six-sided dice is rolled once.

Write down the probability of rolling a 3.

1 There are 6 equally likely outcomes and only one of them is a 3.

2
$$\text{Probability} = \frac{\text{number of successful outcomes}}{\text{total number of outcomes}}$$

Answer: $\frac{1}{6}$

2 [2 marks]

A bag contains 5 red counters, 3 blue counters and 2 green counters.

One counter is taken at random.

(a) Write down the probability that it is blue.

(b) Write down the probability that it is not red.

1 There are $5 + 3 + 2 = 10$ counters altogether.

2 (a) 3 of the 10 are blue, so $P(\text{blue}) = \frac{3}{10}$

3 (b) Not red means blue or green: $3 + 2 = 5$ counters.

4
$$P(\text{not red}) = \frac{5}{10} = \frac{1}{2}$$

Answer: (a) $\frac{3}{10}$ (b) $\frac{1}{2}$

3 [2 marks]

The probability that it rains tomorrow is 0.35

Work out the probability that it does not rain tomorrow.

1 'Rains' and 'does not rain' are the only two possibilities, so their probabilities add to 1.

2 $P(\text{no rain}) = 1 - 0.35$

Answer: 0.65

4

[2 marks]

A spinner has four colours. The table shows some of the probabilities.

Colour	Red	Blue	Green	Yellow
Probability	0.2	0.3	0.15	x

Work out the value of x .

- 1 All the probabilities of a complete set of outcomes must add up to 1.
- 2 $0.2 + 0.3 + 0.15 = 0.65$
- 3 $x = 1 - 0.65 = 0.35$

Answer: $x = 0.35$

5

[3 marks]

- (a) A fair six-sided dice is rolled 300 times. Estimate how many times it will land on 4.
- (b) A biased coin has $P(\text{head}) = 0.4$. It is tossed 250 times. Estimate the number of heads.

- 1 Expected number = probability $\times$ number of trials.
- 2 (a) $P(4) = \frac{1}{6}$, so the estimate is $\frac{1}{6} \times 300 = 50$
- 3 (b) $0.4 \times 250 = 100$
- 4 These are estimates — the real results will vary a little each time.

Answer: (a) 50 (b) 100

6

[3 marks]

60 students travel to school by bus or on foot. The two-way table is incomplete.

	Walk	Bus	Total
Boys	15		35
Girls			
Total	24		60

A student is chosen at random. Work out the probability that the student is a girl who travels by bus.

- 1 Fill in the table using the row and column totals.
- 2 Boys on the bus: $35 - 15 = 20$
- 3 Girls altogether: $60 - 35 = 25$, and girls who walk: $24 - 15 = 9$
- 4 Girls on the bus: $25 - 9 = 16$ (check the bus column: $20 + 16 = 36$ and $24 + 36 = 60$ ✓)
- 5 $P(\text{girl on the bus}) = \frac{16}{60} = \frac{4}{15}$

Answer: $\frac{4}{15}$ (about 0.267)

7

[3 marks]

Two fair six-sided dice are rolled and their scores are added together.

(a) Work out the probability that the total is 7.

(b) Work out the probability that the total is greater than 9.

1 There are $6 \times 6 = 36$ equally likely outcomes.

2 (a) Totals of 7 come from (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) — 6 outcomes.

3
$$P(7) = \frac{6}{36} = \frac{1}{6}$$

4 (b) Greater than 9 means 10, 11 or 12.

5 10: (4,6), (5,5), (6,4) — 3 ways. 11: (5,6), (6,5) — 2 ways. 12: (6,6) — 1 way.

6 That is 6 outcomes, so the probability is $\frac{6}{36} = \frac{1}{6}$

Answer: (a) $\frac{1}{6}$ (b) $\frac{1}{6}$

8

[3 marks]

A drawing pin is dropped 200 times. It lands point up 130 times.

(a) Estimate the probability that the pin lands point up.

(b) The pin is going to be dropped 500 times. Estimate how many times it will land point up.

1 (a) This is experimental probability (relative frequency).

2
$$P(\text{point up}) \approx \frac{130}{200} = 0.65$$

3 (b) Expected number = 0.65×500

4 = 325 times

Answer: (a) 0.65 (b) 325

9

[4 marks]

In a group of 40 students, 22 study French, 18 study German and 9 study both languages.

(a) How many students study neither language?

(b) One student is chosen at random. Work out the probability that they study French but not German.

(c) Work out the probability that they study at least one of the two languages.

1 Start in the middle of the Venn diagram: 9 study both.

2 French only = $22 - 9 = 13$, and German only = $18 - 9 = 9$

3 (a) Inside the circles there are $13 + 9 + 9 = 31$ students, so $40 - 31 = 9$ study neither.

4 (b)
$$P(\text{French only}) = \frac{13}{40}$$

5 (c) At least one language means the 31 students inside the circles.

6
$$P(\text{at least one}) = \frac{31}{40}$$

Answer: (a) 9 (b) $\frac{13}{40}$ (c) $\frac{31}{40}$

10

[4 marks]

A bag contains 4 red counters and 6 blue counters.

A counter is taken at random, its colour recorded, and it is **put back**. A second counter is then taken.

(a) Work out the probability that both counters are red.

(b) Work out the probability that one counter is red and the other is blue.

1 Because the counter is put back, the probabilities are the same both times.

2 $P(\text{red}) = \frac{4}{10} = 0.4$ and $P(\text{blue}) = \frac{6}{10} = 0.6$

3 (a) Multiply along the branches: $0.4 \times 0.4 = 0.16$

4 (b) There are two ways this can happen: red then blue, or blue then red.

5 Red then blue: $0.4 \times 0.6 = 0.24$, and blue then red: $0.6 \times 0.4 = 0.24$

6 Add the two routes: $0.24 + 0.24 = 0.48$

Answer: (a) 0.16 (b) 0.48

Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

A and B are independent events. $P(A) = 0.3$ and $P(B) = 0.5$

Work out $P(A \text{ and } B)$.

1 For independent events you multiply the probabilities.

2 $P(A \text{ and } B) = 0.3 \times 0.5$

Answer: 0.15

2

[2 marks]

A bag contains 5 red and 7 blue counters. One counter is taken and **not** replaced. A second counter is then taken.

The first counter was red. Write down the probability that the second counter is also red.

1 After one red counter is removed there are 4 red counters left.

2 The total is now $12 - 1 = 11$ counters.

Answer: $\frac{4}{11}$

3

[3 marks]

A bag contains 3 green and 5 yellow sweets.

Two sweets are taken at random without replacement.

Work out the probability that both are green.

1 First sweet: $P(\text{green}) = \frac{3}{8}$

2 One green sweet has now gone, so 2 greens remain out of 7 sweets: $P(\text{green}) = \frac{2}{7}$

3 Multiply along the branches: $\frac{3}{8} \times \frac{2}{7} = \frac{6}{56}$

4 Simplify by dividing top and bottom by 2.

Answer: $\frac{3}{28}$

4

[3 marks]

The probability that a machine breaks down on any given day is 0.05, independently of other days.

Work out the probability that the machine breaks down on at least one of three days.

1 'At least one' is much quicker using the complement.

2 $P(\text{no breakdown on one day}) = 1 - 0.05 = 0.95$

3 $P(\text{no breakdown on all three days}) = 0.95^3 = 0.857375$

4 $P(\text{at least one breakdown}) = 1 - 0.857375 = 0.142625$

Answer: 0.142625 (about 0.143)

5

[4 marks]

60% of the students at a school travel in by bus.

Of those who travel by bus, 15% are late. Of those who do not travel by bus, 5% are late.

(a) Work out the probability that a student chosen at random is late.

(b) A student is late. Work out the probability that this student came by bus.

1 Draw a tree: first branch bus (0.6) or not bus (0.4); second branch late or not late.

2 (a) Bus and late: $0.6 \times 0.15 = 0.09$

3 Not bus and late: $0.4 \times 0.05 = 0.02$

4 $P(\text{late}) = 0.09 + 0.02 = 0.11$

5 (b) This is conditional: out of everyone who is late, what fraction came by bus?

6 $P(\text{bus} \mid \text{late}) = \frac{0.09}{0.11} = \frac{9}{11} \approx 0.818$

Answer: (a) 0.11 (b) $\frac{9}{11}$ (about 0.818)

6

[4 marks]

A bag contains 4 white counters and 6 black counters.

Two counters are taken at random without replacement.

Work out the probability that the two counters are the same colour.

1 There are two ways to succeed: both white, or both black. Work them out separately and add.

2 Both white: $\frac{4}{10} \times \frac{3}{9} = \frac{12}{90}$

3 Both black: $\frac{6}{10} \times \frac{5}{9} = \frac{30}{90}$

4 Add: $\frac{12}{90} + \frac{30}{90} = \frac{42}{90}$

5 Simplify by dividing top and bottom by 6.

Answer: $\frac{7}{15}$ (about 0.467)

7

[4 marks]

100 people were asked whether they own a car (set A) and whether they own a bicycle (set B).

45 own a car, 38 own a bicycle and 20 own both.

One person is chosen at random. Work out

(a) $P(A \cup B)$

(b) $P(A' \cap B')$

(c) $P(A | B)$

1 Fill in the Venn diagram: both = 20, car only = $45 - 20 = 25$, bicycle only = $38 - 20 = 18$

2 Neither = $100 - (25 + 20 + 18) = 100 - 63 = 37$

3 (a) $A \cup B$ means car or bicycle or both: $25 + 20 + 18 = 63$, so $P = \frac{63}{100} = 0.63$

4 (b) $A' \cap B'$ means neither, which is the 37 people outside both circles: $P = 0.37$

5 (c) $P(A | B)$ restricts attention to the 38 bicycle owners.

6 Of those, 20 also own a car, so $P(A | B) = \frac{20}{38} = \frac{10}{19}$

Answer: (a) 0.63 (b) 0.37 (c) $\frac{10}{19}$ (about 0.526)

8

[4 marks]

A bag contains n red counters and 5 blue counters.

Two counters are taken at random without replacement.

The probability that both counters are red is $\frac{7}{22}$

(a) Show that $3n^2 - 17n - 28 = 0$

(b) Work out the number of red counters in the bag.

1 (a) There are $n + 5$ counters in total, so $P(\text{first red}) = \frac{n}{n + 5}$

2 After one red is removed: $P(\text{second red}) = \frac{n - 1}{n + 4}$

3 So $\frac{n(n - 1)}{(n + 5)(n + 4)} = \frac{7}{22}$

4 Cross-multiply: $22n(n - 1) = 7(n + 5)(n + 4)$, so $22n^2 - 22n = 7(n^2 + 9n + 20)$

5 $22n^2 - 22n = 7n^2 + 63n + 140$, giving $15n^2 - 85n - 140 = 0$

6 Divide every term by 5: $3n^2 - 17n - 28 = 0$ as required.

7 (b) Factorise: $(3n + 4)(n - 7) = 0$, so $n = -\frac{4}{3}$ or $n = 7$

8 The number of counters must be a positive whole number, so $n = 7$. Check: $\frac{7}{12} \times \frac{6}{11} = \frac{42}{132} = \frac{7}{22}$
✓

Answer: (b) 7 red counters

9

[4 marks]

In a group of 30 students, 18 like maths, 14 like science and 6 like neither subject.

A student who likes maths is chosen at random.

Work out the probability that this student also likes science.

1 First find how many like at least one subject: $30 - 6 = 24$

2 Use the fact that $18 + 14$ counts the 'both' students twice.

3 Number who like both = $18 + 14 - 24 = 8$

4 The question restricts us to the 18 students who like maths.

5 Of those 18, exactly 8 also like science, so the probability is $\frac{8}{18} = \frac{4}{9}$

Answer: $\frac{4}{9}$ (about 0.444)

A box contains 3 red sweets and 7 green sweets.

Three sweets are taken at random without replacement.

Work out the probability that at least one of them is red.

1 'At least one red' covers many cases, so work out the opposite: no reds at all.

2 First sweet green: $\frac{7}{10}$

3 Second sweet green (6 greens left out of 9): $\frac{6}{9}$

4 Third sweet green (5 greens left out of 8): $\frac{5}{8}$

5 $P(\text{no reds}) = \frac{7}{10} \times \frac{6}{9} \times \frac{5}{8} = \frac{210}{720} = \frac{7}{24}$

6 $P(\text{at least one red}) = 1 - \frac{7}{24} = \frac{17}{24}$

Answer: $\frac{17}{24}$ (about 0.708)